

```
In [1]: import pandas as pd          # Step 1: Import Libraries
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
In [2]: df = pd.read_csv('C:/Users/hp/Desktop/accurate_house_price.csv')    #Step 2: Load Data
print(df.head())
print(df.info())
```

	HouseID	Size_sqft	Bedrooms	Bathrooms	Age_Years	Location_Score	\
0	1	1500	3	2	10	7	
1	2	1800	4	3	5	8	
2	3	1200	2	1	20	6	
3	4	2000	4	3	15	9	
4	5	1700	3	2	8	7	

	Has_Garage	Has_Garden	Price
0	1	0	300000
1	1	1	400000
2	0	0	200000
3	1	1	450000
4	1	0	350000

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	HouseID	10 non-null	int64
1	Size_sqft	10 non-null	int64
2	Bedrooms	10 non-null	int64
3	Bathrooms	10 non-null	int64
4	Age_Years	10 non-null	int64
5	Location_Score	10 non-null	int64
6	Has_Garage	10 non-null	int64
7	Has_Garden	10 non-null	int64
8	Price	10 non-null	int64

dtypes: int64(9)

memory usage: 852.0 bytes

None

```
In [3]: print(df.isnull().sum())    #Step 3: Check for Missing Values
```

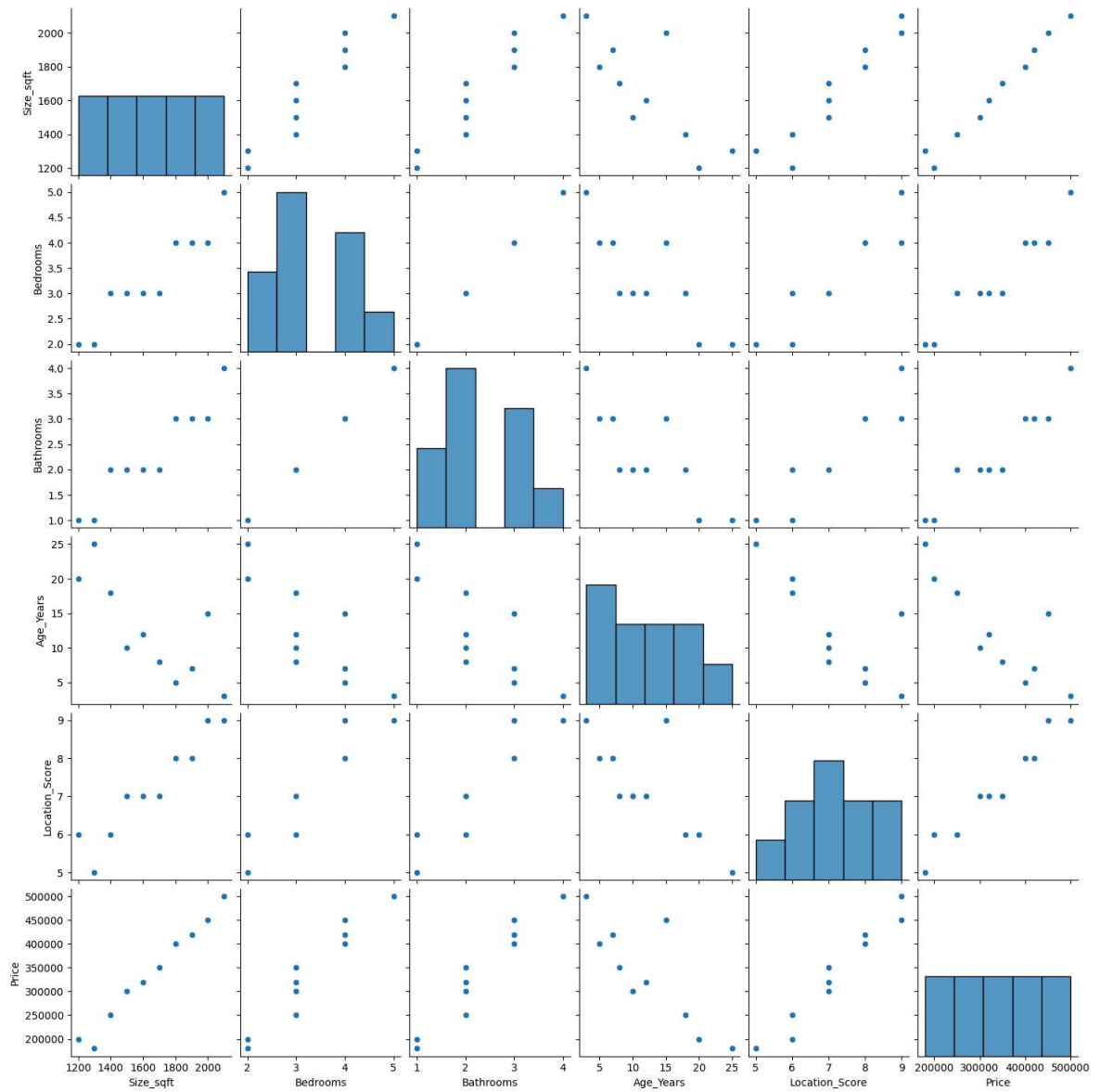
HouseID	0
Size_sqft	0
Bedrooms	0
Bathrooms	0
Age_Years	0
Location_Score	0
Has_Garage	0
Has_Garden	0
Price	0

dtype: int64

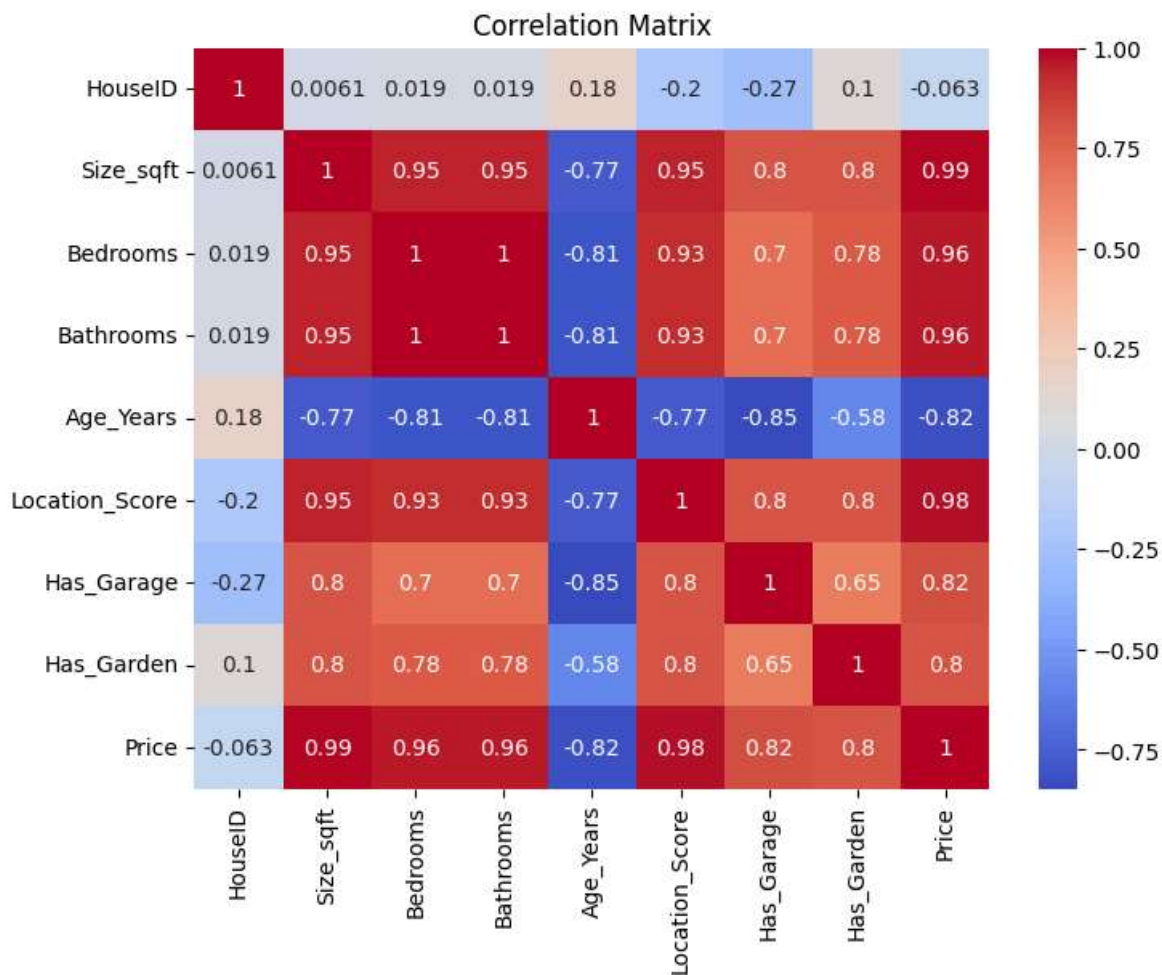
```
In [4]: sns.histplot(df['Price'], kde=True)    # Step 4: Data Exploration & Visualization
plt.title('Price Distribution')                #Check distribution of price:
plt.show()
```



```
In [6]: sns.pairplot(df[['Size_sqft', 'Bedrooms', 'Bathrooms', 'Age_Years', 'Location_Score']])  
plt.show()
```



```
In [7]: plt.figure(figsize=(8,6)) #Correlation heatmap:
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```



```
In [8]: #Step 5: Prepare Data for Modeling
X = df.drop(['HouseID', 'Price'], axis=1) #Feature
y = df['Price'] #Target
```

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In [9]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_
```

```
In [10]: model = LinearRegression() # Step 7: Train Linear Regression Model
model.fit(X_train, y_train)
```

```
Out[10]: LinearRegression
```

```
LinearRegression()
```

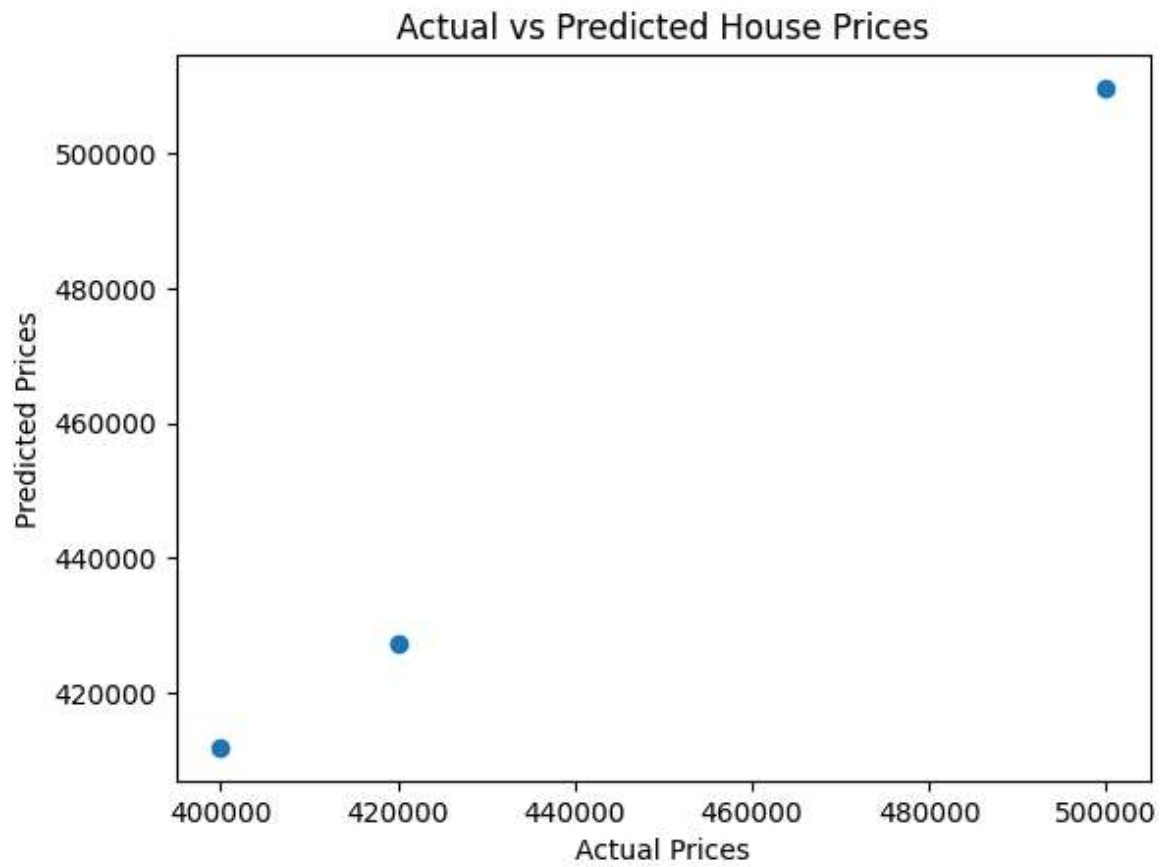
```
In [11]: y_pred = model.predict(X_test) #Step 8: Predict on Test Data
```

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In [12]: mse = mean_squared_error(y_test, y_pred) #Step 9: Evaluate Model
r2 = r2_score(y_test, y_pred)
print(f"Mean Squared Error: {mse}")
print(f"R-squared: {r2}")
```

Mean Squared Error: 96445060.01844412

R-squared: 0.9483330035615478

```
In [13]: plt.scatter(y_test, y_pred) #Step 10: Visualize Actual vs Predicted Pri
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()
```



In []: